

both compilation and execution, and be perfectly effective in writing reliable and robust programs. Go is a statically-typed language with programming syntax similar to C programming. It provides garbage collection, type safety, dynamic typing capabilities and many advanced built-in types like variable length arrays and key value maps.

Go is expressive, concise, clean and efficient. Its concurrency mechanisms make it easy to write programs that get the most out of multi-core and networked machines, while its novel type system enables flexible and modular program construction. Go compiles quickly to machine code, yet has the convenience of garbage collection and the power of runtime reflection. It's a fast, statically typed, compiled language that feels like a dynamically typed, interpreted language.

The latest version of Go is 1.6.3.

Here's the syntax for the 'Hello World' program in Go:

```
package main
import "fmt"
func main() {
    fmt.Println("Hello World")
}
```

The advantages of Go

- Go language is very fast in compilation and execution, compared to other languages, and even the compiler compiles the program really fast. It's a great modern language with high performance.
- Go is regarded as a highly powerful language with built-in concurrency and a high degree of abstraction. It is equipped with Go routines to start concurrent work and the concept of channels to permit both communication and synchronisation.
- Go, being a high level language, has standard documentation features and a powerful rich standard library that contains a full-fledged, working Web server.
- Go has an active community that offers support for any sort of problems.

The disadvantages

- Go, being a new programming language, does not have enough libraries, so developers work hard to develop libraries on their own.
- Go does not have enough resources in terms of books, research articles and other online resources for users to learn in a systematic manner.
- Go is a tough language to learn and error handling is a tedious task.

2. Swift

Swift is a new general-purpose, multi-paradigm, compiled programming language for iOS and OS X, watchOS, tvOS and Linux. It is being built by Apple Inc., on the best of

C and Objective-C, without any of the constraints of C compatibility. The development of Swift was started by Chris Lattner with other programmers at Apple, by taking into consideration ideas from many programming languages like Objective-C, Rust, Haskell, Ruby, Python, C# and CLU. Swift was basically developed to work with Apple's Cocoa and Cocoa Touch frameworks, and has an inbuilt LLVM compiler framework included in Xcode 6, using the Objective-C runtime library.

The latest version of Swift is 3.0 Preview 2.

The syntax for the 'Hello World' program in Swift is:

```
import Cocoa
/* My first program in Swift */
var myString = "Hello, World!"
println(myString)
```

The advantages of Swift

- Swift is comparatively easy to read and write because of less overhead and syntax requirements, and it is the friendliest language for beginners.
- Swift programs are overall shorter in length, which gives programmers unique advantages in terms of passing functions as variables. Swift enables developers to write highly generic code, which can perform lots of different things and reduces repetition.
- Swift compiles directly to native code and utilises both Objective 2.0 runtime as well as Apple's ARC memory management technology.
- Swift is less error-prone as its syntax and language construction exclude several types of mistakes, thereby making it relatively free from crashes and unexpected behaviour.

Swift's disadvantages

- Most of the examples which come inbuilt with Swift are written in Objective C, so to begin Swift programming, one has to learn Objective C.
- Swift is still undergoing a major paradigm shift as newer versions are coming out with lots of changes, which sometimes become hard for professionals and newbies to understand.
- Developers can use Swift only for iOS and OS X app development, so it has limited programming platform opportunities.
- Swift is slower than Objective-C, which means for all modern syntax, simplified code construction, playground app simulation and testing, Swift programming code has a longer execution time period.

3. Hack

Hack is an open source general-purpose programming and scripting language for HipHop Virtual Machine (HHVM) created by Facebook as a dialect for PHP. Hack is specially